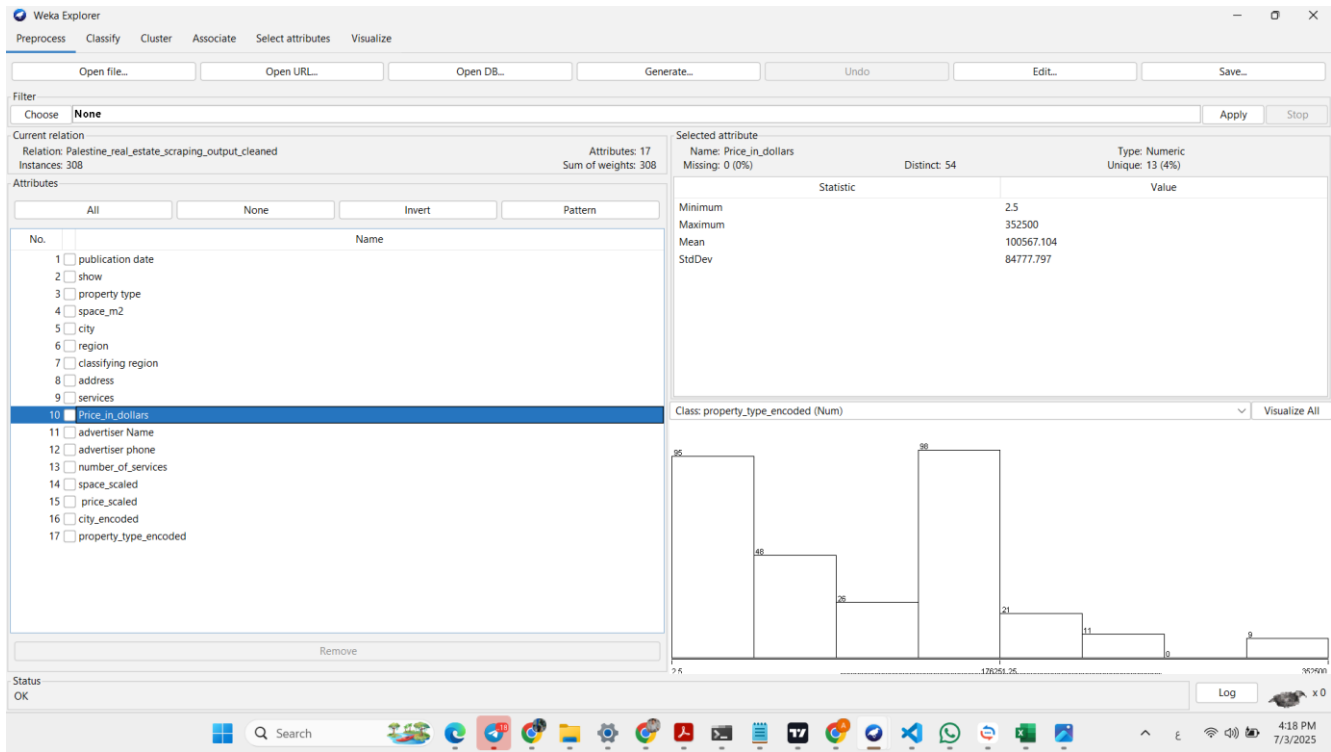
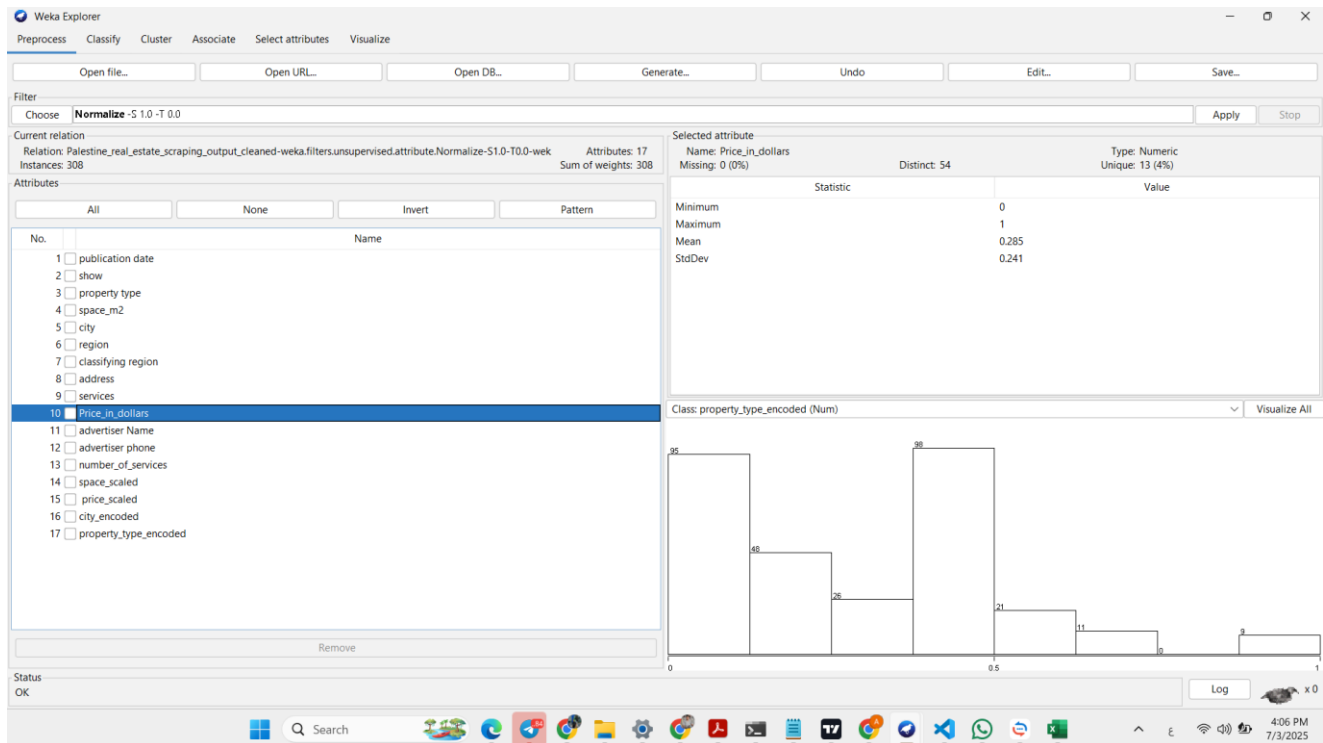


WEKA Tasks

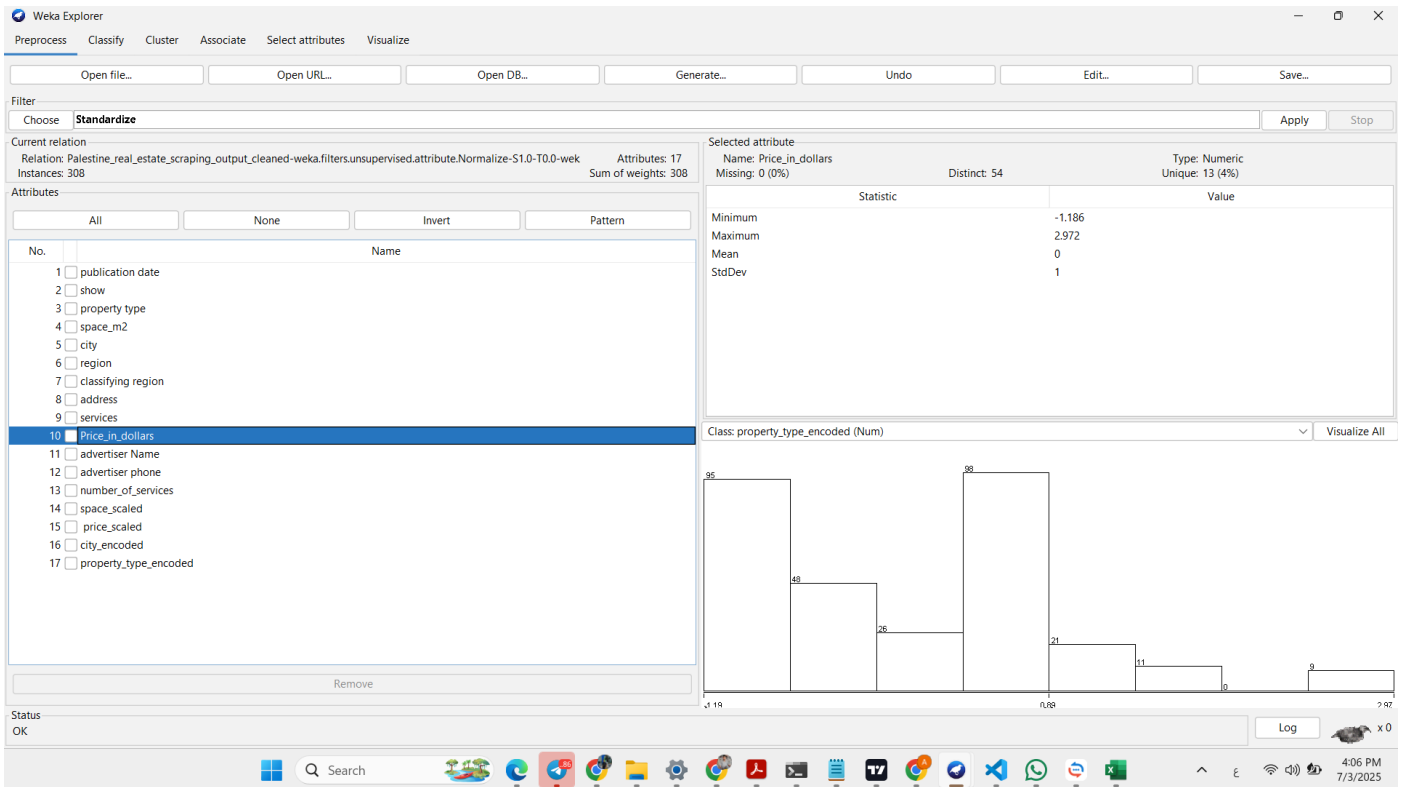
1. قبل ال (Normalization/Standardization)



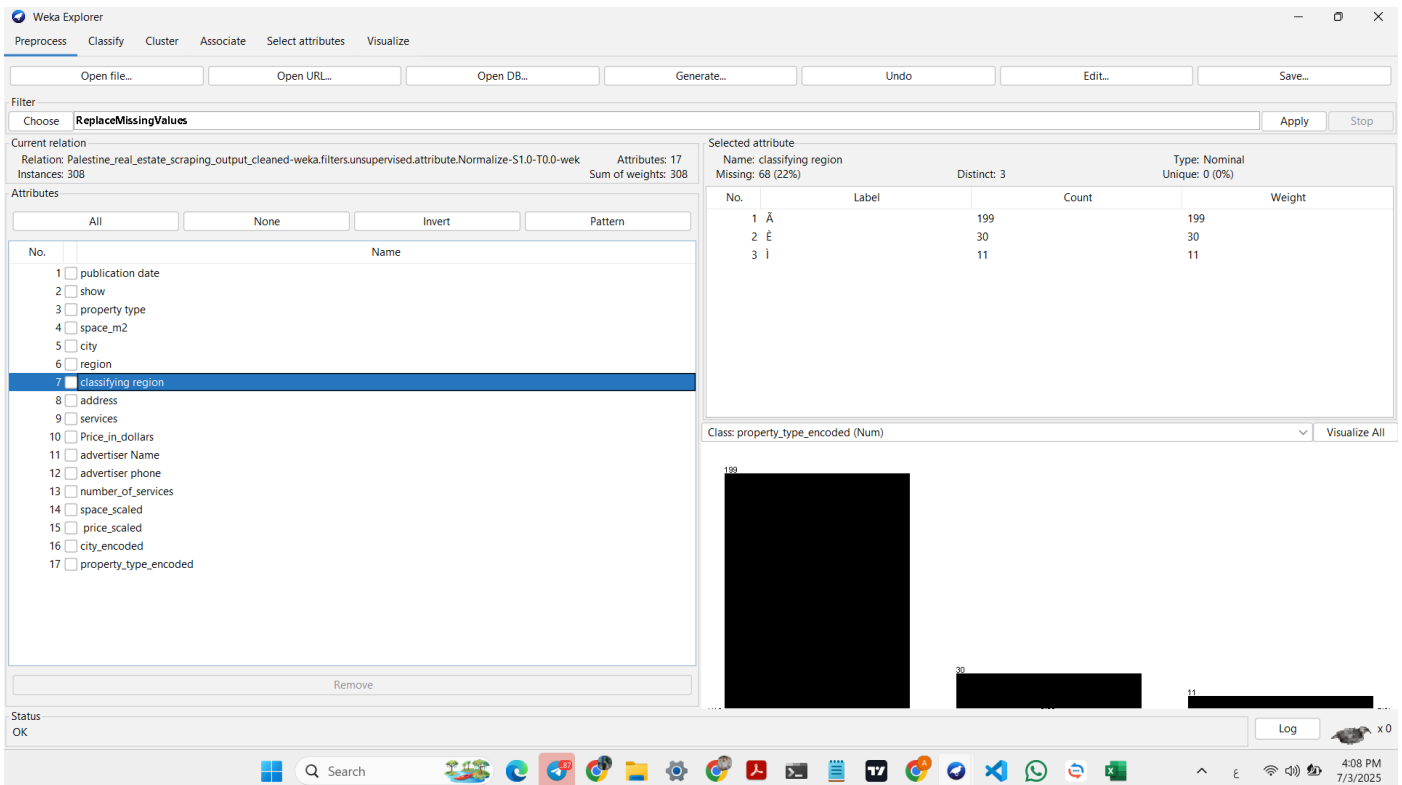
2. بعد ال Normalization



3. بعد ال Standardization



4. قبل ال ReplaceMissingValues



5. بعد ال ReplaceMissingValues

The screenshot shows the Weka Explorer interface with the 'Preprocess' tab selected. The 'ReplaceMissingValues' filter is applied to the 'classifying region' attribute. The 'Attributes' list on the left shows 17 attributes, with 'classifying region' selected. The 'Selected attribute' panel on the right shows the 'classifying region' attribute with 3 distinct values: 'A', 'E', and 'I'. The 'Class: property_type_encoded (Num)' is selected at the bottom.

No.	Label	Count	Weight
1	A	267	267
2	E	30	30
3	I	11	11

6. السمات الأكثر أهمية Feature Selection

The screenshot shows the Weka Explorer interface with the 'Select attributes' tab selected. The 'Ranker' filter is applied to the 'property_type_encoded' attribute. The 'Attribute selection output' panel shows the results of the feature selection process, including the ranked attributes and the selected attributes.

Attribute Selection Mode: ☒ Use full training set

Attribute selection output:

Search Method: Attribute ranking.

Attribute Evaluator (supervised, Class (numeric): 17 property_type_encoded): Classifier feature evaluator

Using Wrapper Subset Evaluator

Learning scheme: weka.classifiers.rules.ZeroR

Scheme options:

Subset evaluation: RMSE

Number of folds for accuracy estimation: 5

Ranked attributes:

- 0 16 city_encoded
- 0 15 price_scaled
- 0 6 region
- 0 5 city
- 0 4 space_m2
- 0 3 property_type
- 0 2 show
- 0 7 classifying region
- 0 8 address
- 0 9 services
- 0 13 number_of_services
- 0 14 space_scaled
- 0 12 advertiser phone
- 0 10 Price_in_dollars
- 0 11 advertiser Name
- 0 1 publication date

Selected attributes: 16,15,6,5,4,3,2,7,8,9,13,14,12,10,11,1 : 16

Random Forest .7

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'RandomForest' with parameters: -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1. The test options are set to 'Cross-validation' with 10 folds. The result list on the left shows several entries, with '16:10:24 - trees.RandomForest' selected. The classifier output pane displays the following text:

```
space_m2
city
region
classifying region
address
services
Price_in_dollars
advertiser Name
advertiser phone
number_of_services
space_scaled
price_scaled
city_encoded
property_type_encoded
Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

RandomForest

Bagging with 100 iterations and base learner

weka.classifiers.trees.RandomTree -K 0 -M 1.0 -V 0.001 -S 1 -do-not-check-capabilities

Time taken to build model: 0.08 seconds

=== Cross-validation ===
=== Summary ===

Correlation coefficient      0.9532
Mean absolute error         0.983
Root mean squared error     1.2084
Relative absolute error     55.8401 %
Root relative squared error 56.6731 %
Total Number of Instances   308
```

The status bar at the bottom indicates 'OK' and the system clock shows 4:10 PM on 7/3/2025.

SMOreg .8

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'SMOreg' with parameters: -C 1.0 -N 0 -I "weka.classifiers.functions.supportVector.RegSMOImproved -T 0.001 -V -P 1.0E-12 -L 0.001 -W 1" -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007". The test options are set to 'Cross-validation' with 10 folds. The result list on the left shows several entries, with '16:11:36 - functions.SMOreg' selected. The classifier output pane displays the following text:

```
+ 0.001 * (normalized) advertiser phone=598253577
+ 0.0005 * (normalized) advertiser phone=597590066
+ 0 * (normalized) advertiser phone=593151934
+ 0.009 * (normalized) advertiser phone=592108731
+ 0.006 * (normalized) advertiser phone=598532915
+ 0.0052 * (normalized) advertiser phone=599118460
- 0.0054 * (normalized) advertiser phone=595909993
+ 0.0113 * (normalized) advertiser phone=585600444
- 0.0104 * (normalized) advertiser phone=569378999
+ 0.0184 * (normalized) advertiser phone=596989899
- 0.0116 * (normalized) advertiser phone=522980275
+ 0.007 * (normalized) advertiser phone=569150444
+ 0.0049 * (normalized) advertiser phone=584321123
- 0.0028 * (normalized) advertiser phone=597651181
- 0.0178 * (normalized) number_of_services
- 0.0481 * (normalized) space_scaled
+ 0.0089 * (normalized) price_scaled
- 0.0039 * (normalized) city_encoded
+ 0.5006

Number of kernel evaluations: 47586 (93.246% cached)

Time taken to build model: 0.13 seconds

=== Cross-validation ===
=== Summary ===

Correlation coefficient      0.9888
Mean absolute error         0.2718
Root mean squared error     0.3639
Relative absolute error     15.4394 %
Root relative squared error 17.067 %
Total Number of Instances   308
```

The status bar at the bottom indicates 'OK' and the system clock shows 4:11 PM on 7/3/2025.

